

CS 229 Final Exam Cheatsheet

Ver Steeg · Q1-11 Conceptual | Q12-16 HW | Q17-20 Calculation · HW3 & GNNs NOT on exam

1 · PROBABILITY & SOFTMAX [Q1, Q0]

$$\text{softmax}(z_{\blacksquare}) = \exp(z_{\blacksquare}) / \sum \exp(z_{\blacksquare})$$

Logits $(-\log 2, +\log 2) \rightarrow (1/5, 4/5)$ $(0, \log 3) \rightarrow (1/4, 3/4)$

Temp T: divide logits by T. $T \rightarrow 0 = \text{argmax}$. $T \rightarrow \infty = \text{uniform}$.

$$\text{Bayes: } p(\theta|D) \propto p(D|\theta) \cdot p(\theta) \quad \text{MAP} = \text{argmax posterior}$$

$$\text{Marginal: } P(X=0) = \sum_{\blacksquare} P(X=0, Y=y) \quad \text{Cond: } P(Y|X) = P(X, Y)/P(X)$$

PDF: $p(z) \geq 0$, integrates to 1. CAN exceed 1 (unlike PMF).

2 · PYTORCH BASICS [Q0] ■ missed

■ $A*B$ = element-wise. ■ $A@B$ = matrix multiply.

`loss.backward()` → fills `.grad` only. Does NOT: update params, zero grads, free memory.

Order: forward → loss → `zero_grad()` → `backward()` → `step()`

3 · MLP ARCHITECTURE [Q2, HW0]

`nn.Linear(in_dim, out_dim)` — first arg = input, second = output

HW0 NameMLP: Embedding(27,16) → **Flatten** → Linear(240,128) → ReLU → Linear(128,64) → ReLU → Linear(64,2)

HW0 tokenize: `ord(c)`–96. `a=1`, `z=26`, `pad=0`. `tokenize("dog")`=[4,15,7,0,0]

4 · SELF-ATTENTION [Q1]

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \cdot V$$

Complexity: $O(N^2 \cdot d)$ — N = sequence length

■ The $1/\sqrt{d}$ factor is a TEMPERATURE (prevents saturation at init). It does NOT normalize softmax output.

Without positional embeddings: permutation-invariant, can't distinguish word order.

RoPE: rotate Q,K so dot product depends on relative position. Used in Qwen/LLaMA.

NoPE: no positional embedding (HW3 ViT). Works because $\|z\|$ encodes t in high-d.

5 · BERT vs GPT [Q1] ■ missed

	GPT	BERT
Style	Decoder, causal	Encoder, bidirectional
Pretraining	Next-token prediction	Masked LM (MLM)
P(seq)?	YES — autoregressive	NO
MLM	—	15%: 80%[MASK] 10%rand 10%unch

■ Only GPT gives a probability distribution over sequences.

6 · LR vs BATCH SIZE [Q4] ■ missed

$$\text{noise} \propto \text{LR} / \text{batch_size} \Rightarrow \text{new_LR} = \text{old_LR} \times (\text{B_new} / \text{B_old})$$

■ $4 \times \text{batch size} \rightarrow 4 \times \text{LR}$ (NOT divide by 4).

`batch=10`, `lr=0.001` on 1 GPU → `batch=40` on 4 GPUs → `lr=0.004`

7 · MOMENTUM & ADAM [Q4]

$$\text{Momentum: } v_{\blacksquare} = \mu \cdot v_{\blacksquare} + \eta \cdot g_{\blacksquare} \quad \theta = \theta - v_{\blacksquare}$$

Requires storing $v_{\blacksquare}$. No extra storage for vanilla GD.

$$\text{Adam } m_{\blacksquare} = \beta m_{\blacksquare} + (1-\beta) g_{\blacksquare} \quad (\text{EMA of gradients})$$

$$\text{Adam } v_{\blacksquare} = \beta v_{\blacksquare} + (1-\beta) g_{\blacksquare}^2 \quad (\text{EMA of squared grads})$$

$$\text{Update: } \theta = \theta - \eta / (\sqrt{v_{\blacksquare}} + \epsilon) \cdot m_{\blacksquare}$$

Adaptive in: (A) changes over time, (B) different per parameter.

8 · LoRA [Q2, HW1]

$$\Delta W = B \cdot A \quad A \in \mathbb{R}^{(r \times d_{\text{in}})} \quad B \in \mathbb{R}^{(d_{\text{out}} \times r)}$$

$$\text{Trainable params} = r(d_{\text{in}} + d_{\text{out}}) \quad \blacksquare \quad d_{\text{in}} \times d_{\text{out}}$$

Rank-1 matrix = outer product of two vectors (all rows proportional).

QLoRA: quantize frozen base to 4-bit, LoRA adapters stay float16.

HW1 trainable: 1,081,344 / 495,114,112 = 0.22% of params.

9 · VAE vs GAN [Conceptual]

	VAE	GAN
Prob. model	YES (ELBO)	NO
Encoder	$q\phi(z x)$	None
Training	Maximize ELBO	Adversarial minimax
Output	Blurry, stable	Sharp, unstable
Latent	Continuous, structured	Unstructured noise

$$\text{ELBO} = E[\log p(x|z)] - \text{KL}(q\phi(z|x) || p(z))$$

$$\text{GAN: } \min_{\blacksquare} \max_{\blacksquare} E[\log D(x)] + E[\log(1-D(G(z)))]$$

10 · CALIBRATION & ECE [Q3, HW2] ■ missed Q4+Q5

$$\text{ECE} = \sum_b (n_b/N) \cdot | \text{acc}(b) - \text{conf}(b) |$$

Overconfident: `conf > acc`. Underconfident: `conf < acc`.

■ 80% conf, 5/5 rain → `acc=100%`, `conf=80%` → `ECE=0.2`, UNDERCONFIDENT

■ Binary ECE: use `P(positive)` as confidence (allows full [0,1], not [0.5,1])

Method	ECE	When
Base model	0.1278	—
MC Dropout (30×)	0.1022	Inference
Label Smoothing	0.0397	Train
★ Temp Scaling T=3	0.0373	Post-hoc

11 · InfoNCE LOSS [Q3, CALC Q18] ■ missed

$$L = -\log \left[\exp(z_{\text{a},z_{\text{p}}}) / (\exp(z_{\text{a},z_{\text{p}}}) + \sum_k \exp(z_{\text{a},z_{\text{k}}})) \right]$$

Anchor z_{a} , positive z_{p} (same image, diff aug), negatives z_{k} (other images).

Minimize = push anchor toward positive, away from all negatives.

CLIP: positive = (image, caption). Zero-shot = cosine similarity to class names.

DINO: teacher = EMA of student. Student predicts teacher pseudo-labels.

12 · CONVOLUTION [Q2, CALC Q20]

valid: $|y| = |x| - |h| + 1$ same: output size = input size

Example: `x`=[3,1,4,1,5], `h`=[1,0,-1]. Output: [-1, 0, -1]

2D: input (32,32,3), 3×3 kernel, same, 4 channels → (32,32,4).

1×1 conv: linear transform across channels per pixel. No spatial mixing.

FCN: no FC layers → any input size. FC layers break variable size.

Property	Classification	Segmentation
Rotation	Invariance (output same)	Equivariance (output rotates)

13 · ATTENTION CALC [CALC Q17]

$x_{\blacksquare}$ =[1,0], $x_{\blacksquare}$ =[0,1], $x_{\blacksquare}$ =[1,1]. $K=Q=V=\text{identity}$. Query = $x_{\blacksquare}$.

Dot products: $x_{\blacksquare} \cdot x_{\blacksquare}=1$, $x_{\blacksquare} \cdot x_{\blacksquare}=0$, $x_{\blacksquare} \cdot x_{\blacksquare}=1$

$$\text{softmax}([1,0,1]) \approx [0.42, 0.16, 0.42]$$

$$\text{Output} = 0.42x_{\blacksquare} + 0.16x_{\blacksquare} + 0.42x_{\blacksquare} = [0.84, 0.58]$$

$x_{\blacksquare}$ and $x_{\blacksquare}$ tied for highest attention (same dot product with query).

14 · HW1 PROMPTING [Q14]

Method	Overall	Lvl 1	Lvl 2	Lvl 3
Base	4.7%	10%	2%	2%
System	11.3%	30%	2%	2%
CoT	16.0%	38%	6%	4%
ICL	10.7%	28%	2%	2%
LoRA	36.7%	96%	10%	4%

ICL: label space + format drives perf — not whether example labels are correct.

15 · SCALING LAWS & GOODHART

Test loss $\propto N^{\alpha}$ (power law in model params N and data D).

Chinchilla: scale N and D equally. Most models undertrained. “Optimal = underfitting.”

Goodhart: “When a measure becomes a target, it ceases to be a good measure.”